



PHM North America 2024 Challenge (Group A1)

Preventive Maintenance For Robotics And Intelligent Automation Course
Examination Project

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PHM North America 2024 Challenge



The challenge

What is the challenge we're about to take on?



SOTA

What strategies have already been developed?



Our proposal

What we expect to do?

The goal of the challenge

- ❑ The **goal** of this challenge is to assess the **health of helicopter turbine engines** using machine learning models.
- ❑ Training data includes four engines, but their observations are shuffled, and asset IDs are removed. The remaining three engines are used for testing and validation, requiring models to **generalize**.
- ❑ Engine health is assessed by looking at the **torque margin** as an indicator of engine health:

$$trq_margin = \frac{trq_measured - trq_target}{trq_target} \times 100$$

- ❑ The **regression goal**: estimating the torque margin expressed at a *probability distribution function* (PDF).
- ❑ The **classification goal**: estimating the *binary classification* of the health state (0 = nominal, 1 = faulty), with a *confidence metric*

Dataset features	
Variable	Description
Mean gas temperature (<i>mgt</i>)	Average temperature inside the turbine (°C)
Pressure altitude (<i>pa</i>)	Height above the standard atmospheric pressure level (<i>feet</i>)
Indicated airspeed (<i>ias</i>)	Speed of the helicopter relative to the surrounding air (<i>knots</i>)
Compressor speed (<i>ng</i>)	Speed of the engine's compressor (%)
Torque measured (<i>Tmeasured</i>)	Actual engine torque output (%)
Net power (<i>np</i>)	Effective power output of the engine (%)
Outside air temperature (<i>oat</i>)	Temperature of the external air (°C)

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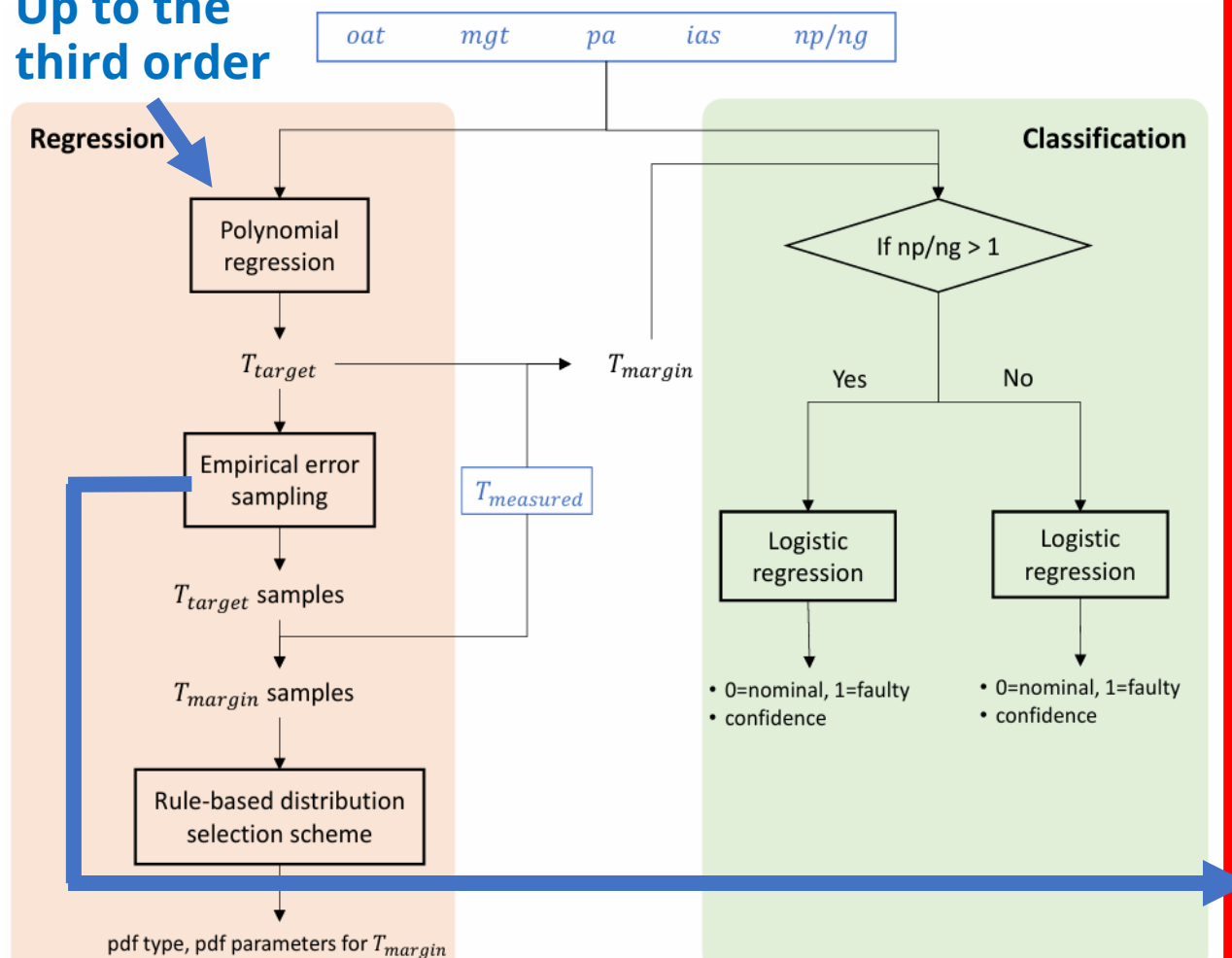


Our proposal

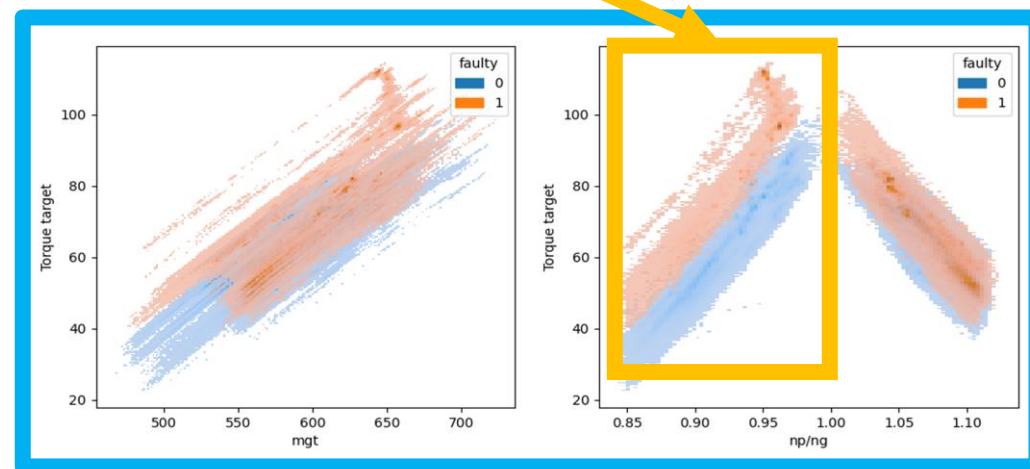
What we expect to do?

A Simple Probabilistic Approach

Up to the third order

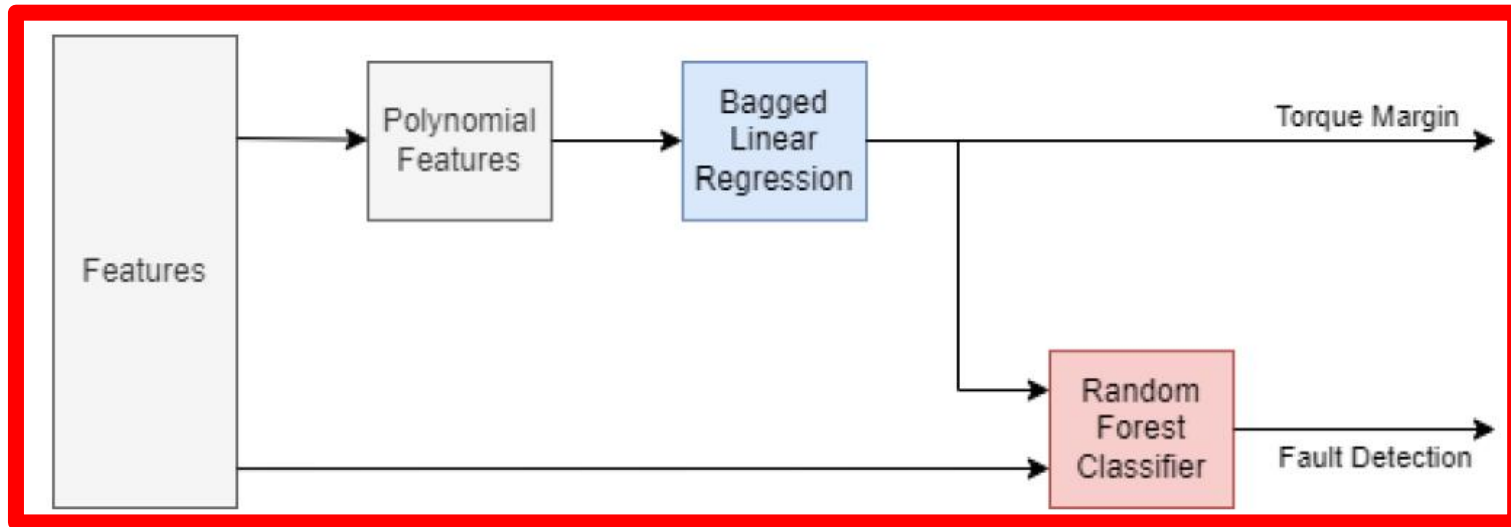


- ❑ The ratio np/ng is used as a feature. It measures *how efficiently the engine converts compressor speed into power*.
- ❑ When $np/ng < 1$, the nominal and faulty cases *become linearly separable* $\Rightarrow$ **2-model classifier**.



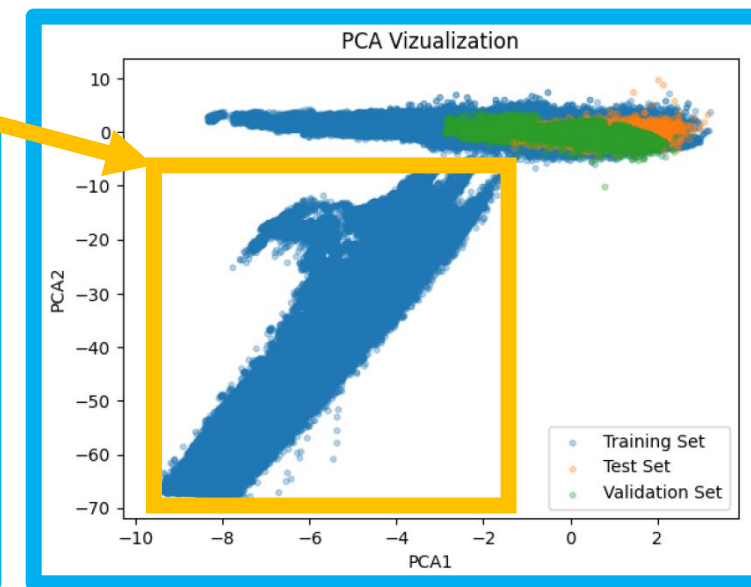
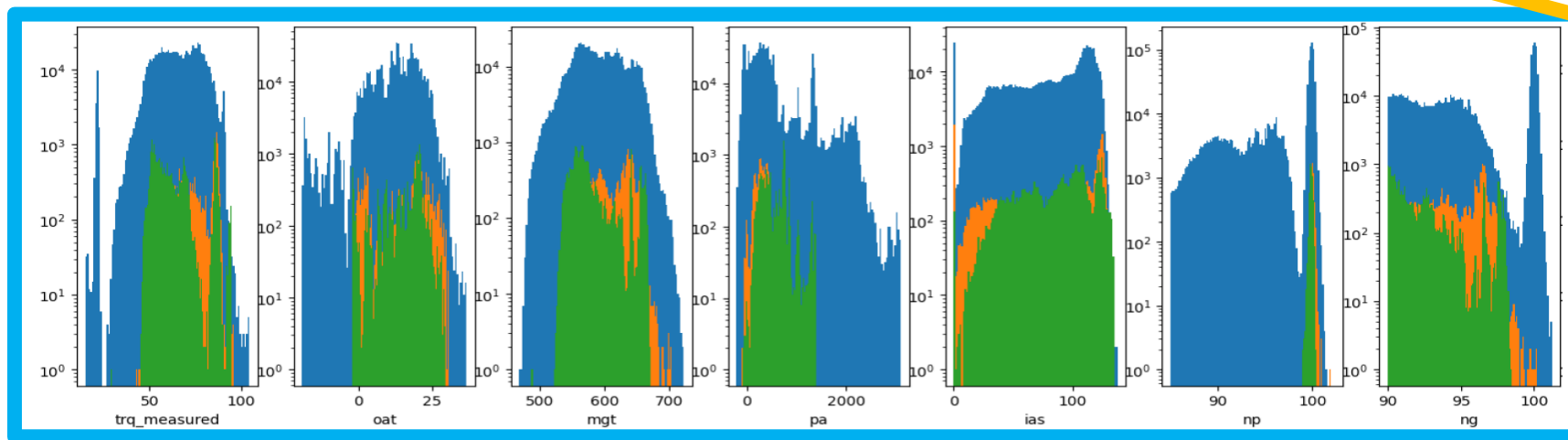
- ❑ A probabilistic behavior is added by adding a random item from the **set of training residuals**.
- ❑ Repeating this process 1.000 times, we obtain an ensemble of prediction that **reflects the PDF**.

A Design Science Approach Comparing Ensemble Learning and Artificial Neural Networks



- ❑ Adding **2°order polynomial features** significantly *boosted the accuracy*
- ❑ The **predicted torque margin** from the regression model is used as a *feature for classification model*
- ❑ **Group C.V.** outperformed K-Fold C.V. using *Mini-batch K-means clustering* to separate groups in the training data

- ❑ PCA is used to **cut a big part of training set** that is not found in the validation and test sets



Estimating with the relationship between torque margin and density altitude

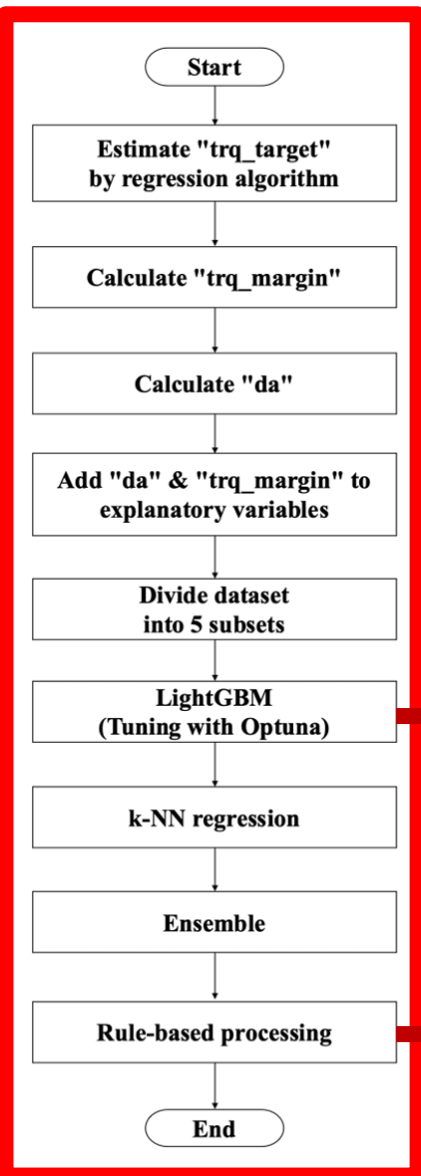


- The **density altitude** is estimated as a linear regression approximation of *pressure altitude* and *air temperature*. $da = 1.2376 \times pa + 118.8 \times oat - 1782$

Density altitude is *the height above ground at which the air is as thick as it is at the current*.

- **Hp1**: The *oat* and *pa* change smoothly over time.
- **Hp2**: Faulty engines tend to have a small *Tm*. Healthy ones also have small *Tm* under high *da*.

- The target torque is estimated by means of a **second-order polynomial regression**
- **Hp1** justifies the **introduction of the k-NN smoothing**. The temporal information is rebuilt by means of *Euclidean distance* in the feature space.
- **Hp2**, proven to hold only on a subset of the training set, justifies the **introduction of a rule-based processing downstream** to the ensemble prediction.



→ A *leaf-wise* growth variant of the *level-wise* growth *XGBoost*

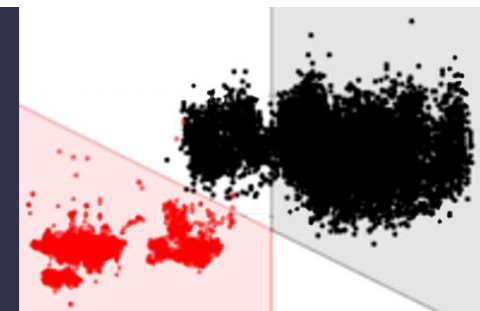
→ If $da < 500$ and $tm < -0.01 \times da + 2.5$, the engine is faulty. Else, it is healthy.



LightGBM

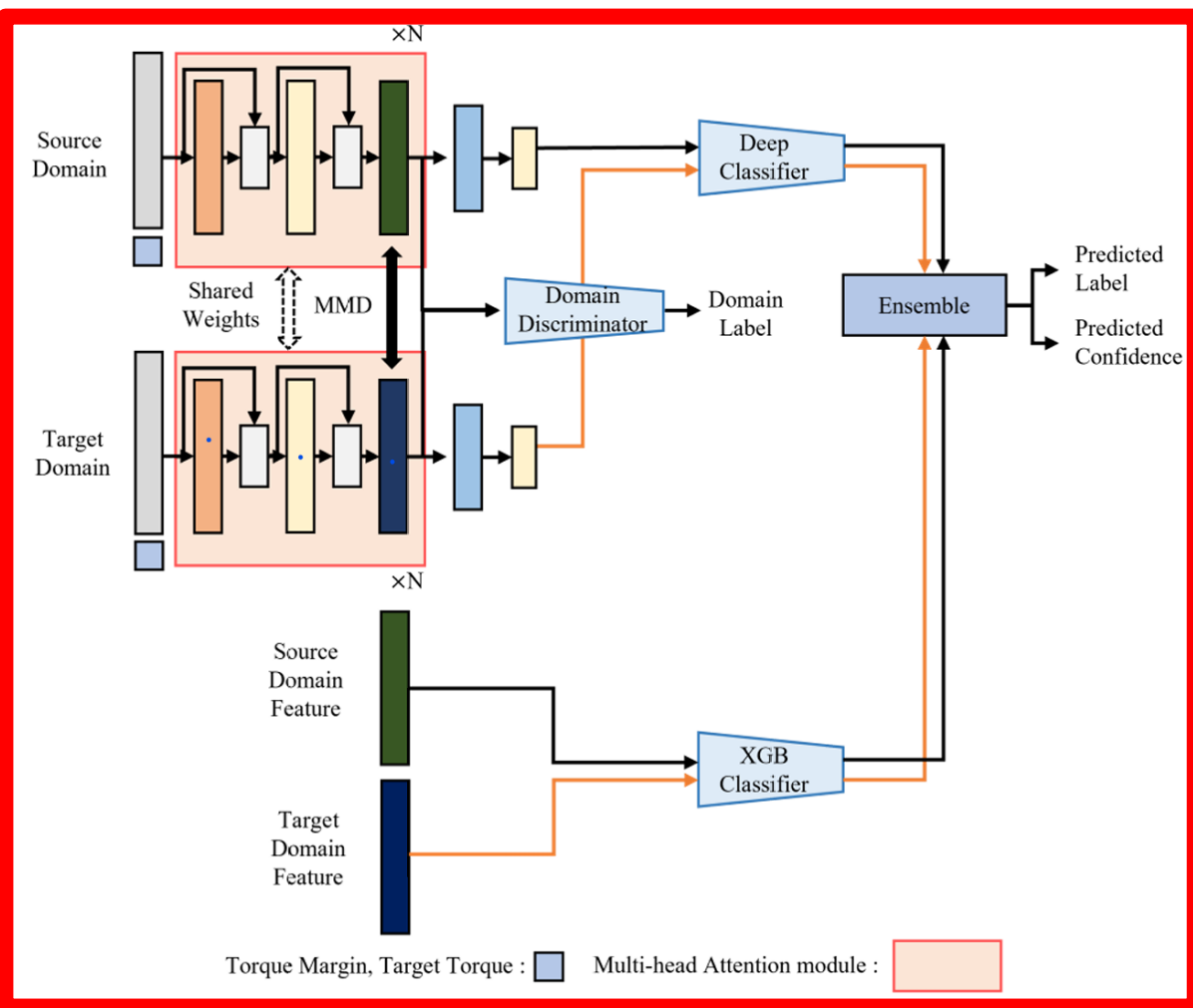


LightGBM+kNN



LightGBM+kNN+rule

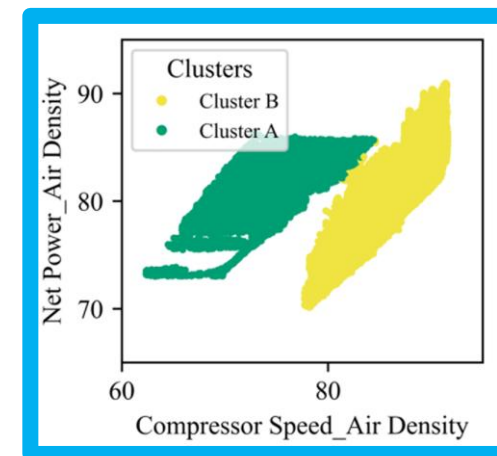
Multi-Head Attention Approach



$$P = P_0 \times \left(1 - \frac{L \cdot h}{T_0} \right)^{\frac{g \cdot M}{R \cdot L}}$$

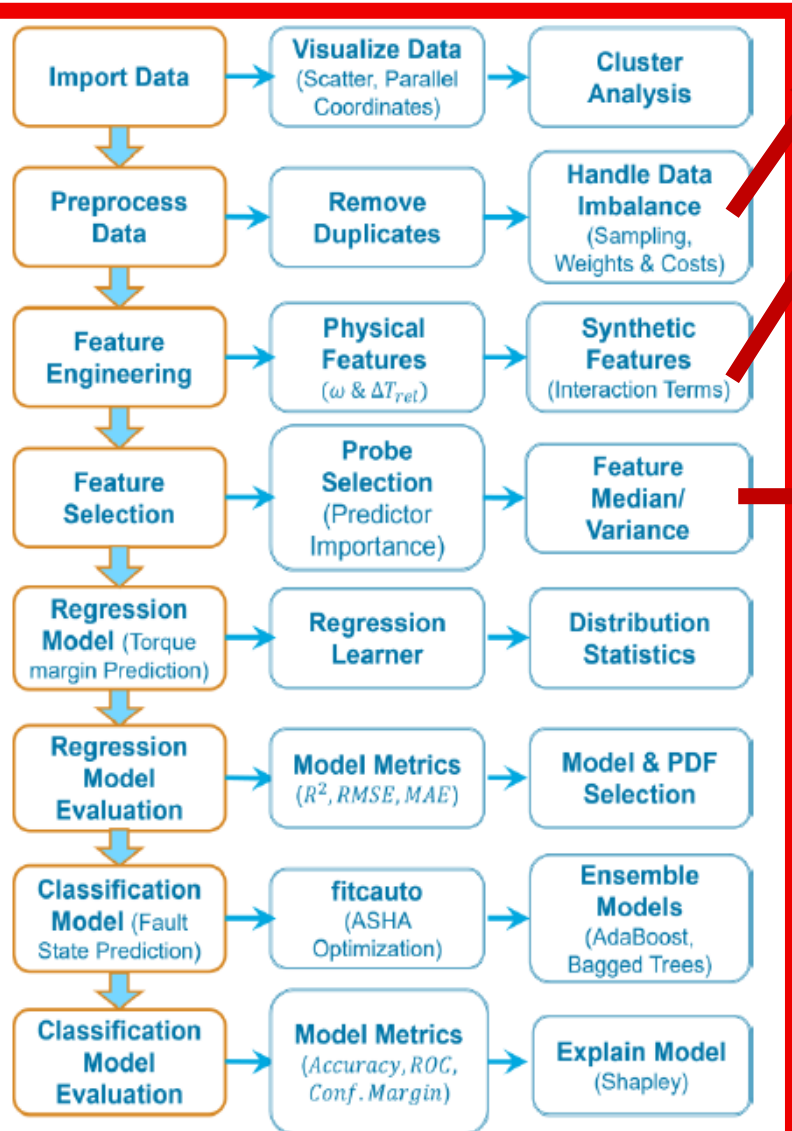
$$\rho = \frac{P \cdot M}{R \cdot T}$$

- Two identical engines might operate differently because one is in *denser* air and the other in *thinner* air $\Rightarrow$ each feature is divided by ρ . (Normalization)
- Gaussian Mixture Model, with shared covariance structure, identifies **2 distinct operational regimes**.
- Cluster A doesn't show in the test and validation sets $\Rightarrow$ it can be discarded.
- A Shared Multi-head Attention is used to **extract complex patterns between the features**, coming from both the training and test sets. A *discriminator* is used to "confuse" about the source of samples.



- An **ensemble** yields the prediction based on both the deep learning and *XGBoost* classifiers.

A Comprehensive Approach with Adaboost Ensemble Model

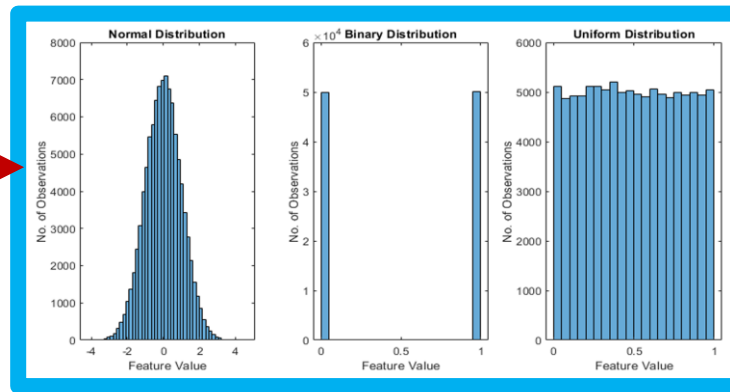


❑ Duplicates were removed and the minority class was upsampled.

$$\Delta T_{relative} = (mgt - oat)/mgt$$

$$\omega = np/Torque_{measured}$$

❑ As the engines age, the relative temperature drops, and the turbine generates less work.



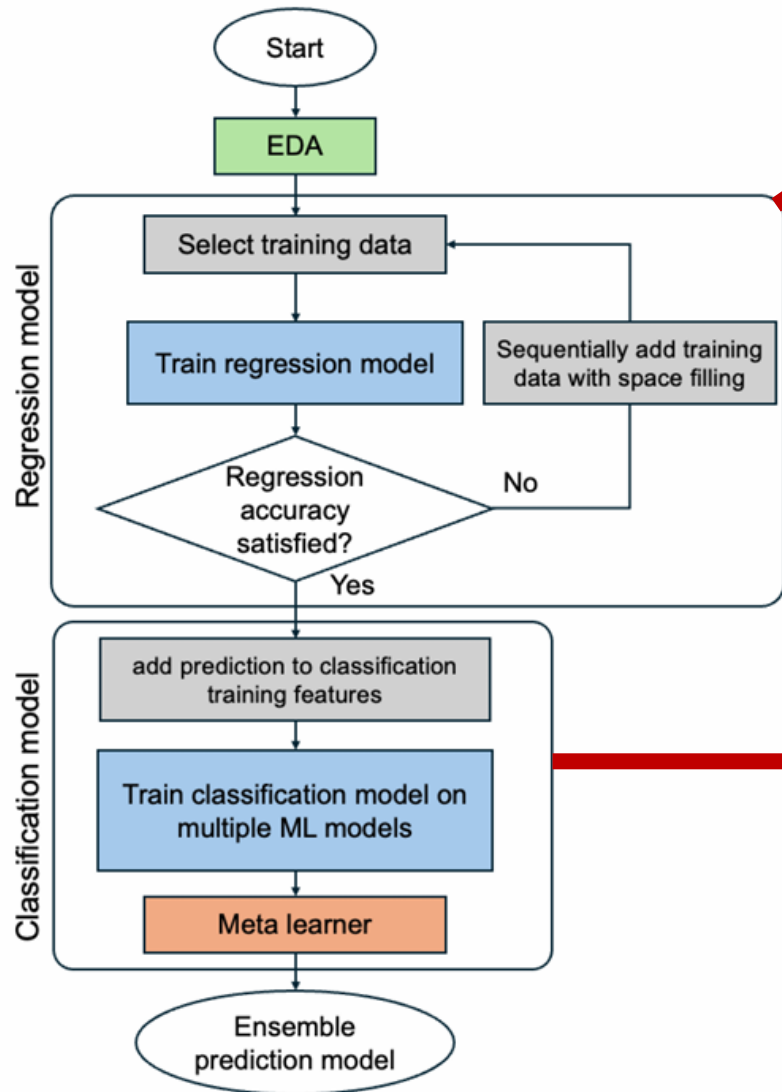
❑ 3 random features sampled from different PDF are added; any original feature with an importance lower than their average is pruned.

❑ A *Parallel Coordinates Plot* is also used for the very same purpose.

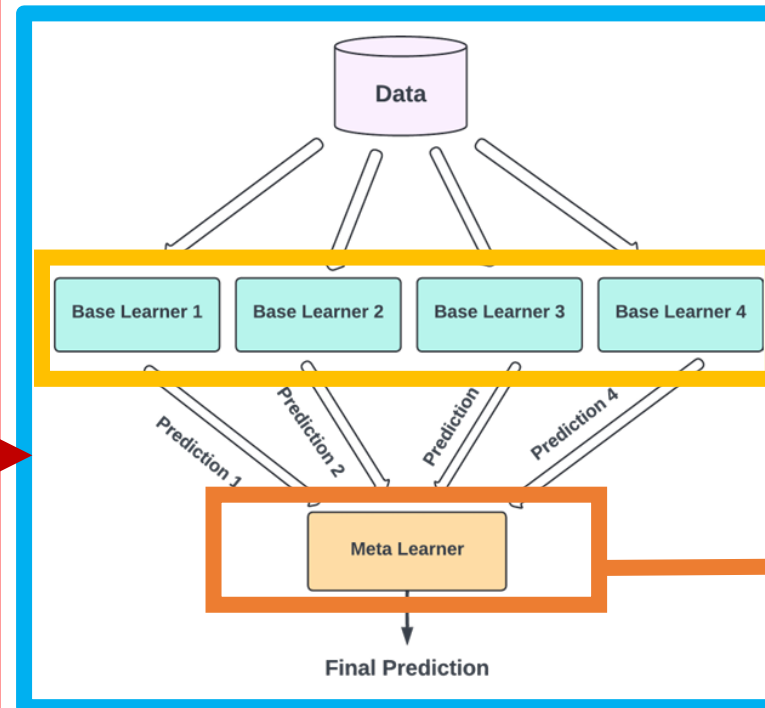
❑ To choose the best model, MATLAB's "fitcauto" function is used, which tested a variety of models on a fraction of datasets. Models that had low validation loss were tested on the whole dataset.

❑ **Adaptive Boosting Trees Ensemble has been preferred** over Bagged Trees Ensemble.

Health Condition Using Ensemble ML Models

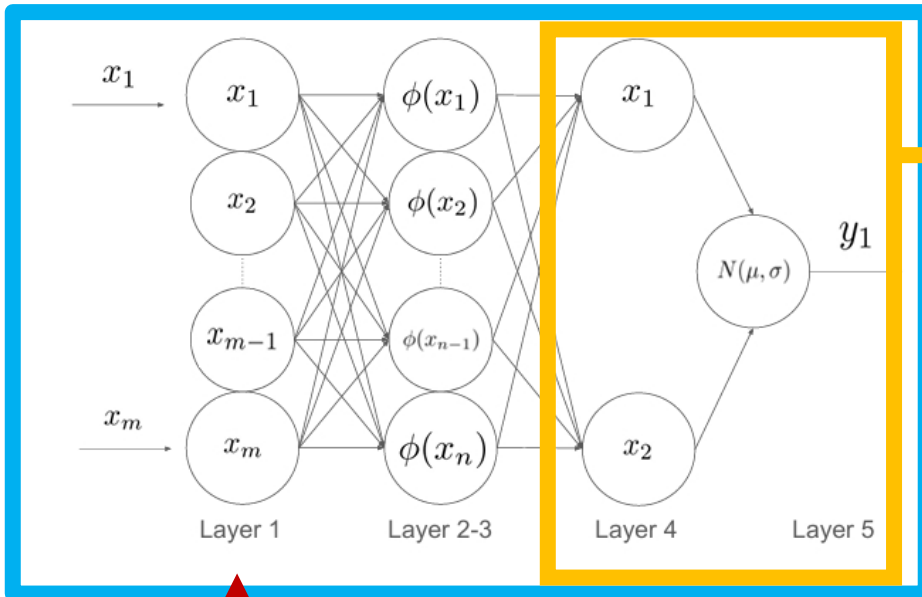


- ❑ **Gaussian Process Regression** is used to estimate the torque margin. Due to its complexity, only a subset of the training set, determined with a **Space-Filling (max-min) strategy**, is fed.



- ❑ An **ensemble model is used for classification** with *Logistic Regression as meta-learner*.
- ❑ CNN, MLP, XGBoost, and AdaBoost — **each make their own independent predictions** about whether the engine is healthy or faulty.
- ❑ The logistic regressor takes the predictions, whether binary or confidence, and **combines them linearly** to produce a probability.

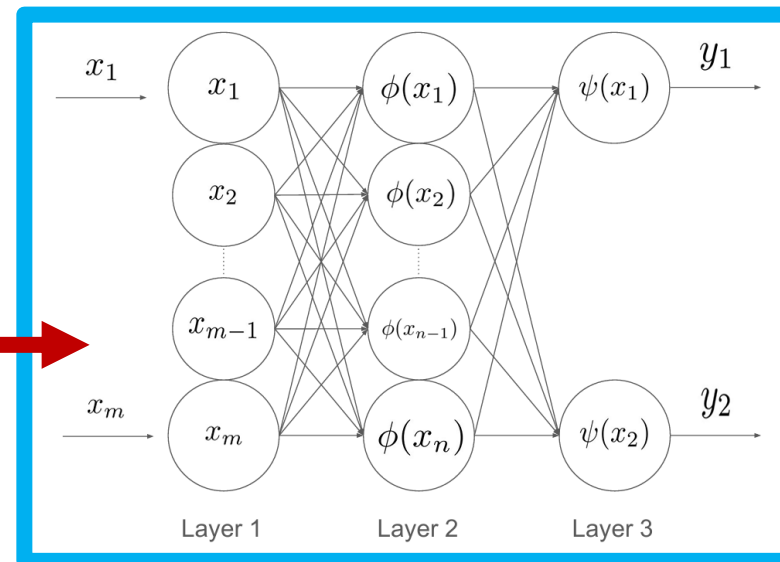
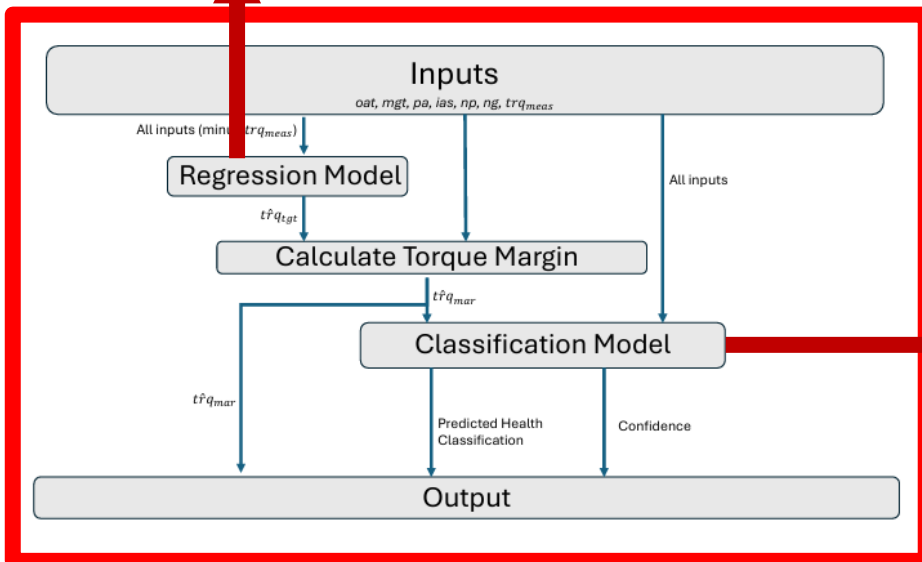
Probabilistic Neural Network



❑ A $[7, 256, 256, 2, 1]$ **MLP architecture with probabilistic end layer** is used for the regression task. Rather than predicting a single value, **the network learns a function that maps each set of input features to its own probability distribution.**

❑ The choice of normal distribution is justified by the *Central Limit Theorem*, which makes it possible to **consider the noises as coming from a normal distribution.**

Negative Log-Likelihood is chosen over classic MSE because the variance of the PDF is not posed as constant. The loss is given by $-\log(f(\bar{y}))$, where $f \sim \mathcal{N}(\mu, \sigma^2)$



❑ A shallow $[7, 32, 2]$ MLP, with σ activations and *softmax* output, is chosen as the binary classifier.

❑ In accordance with the **Occam's Razor principle**, this simple network proved to generalize better than a deeper one

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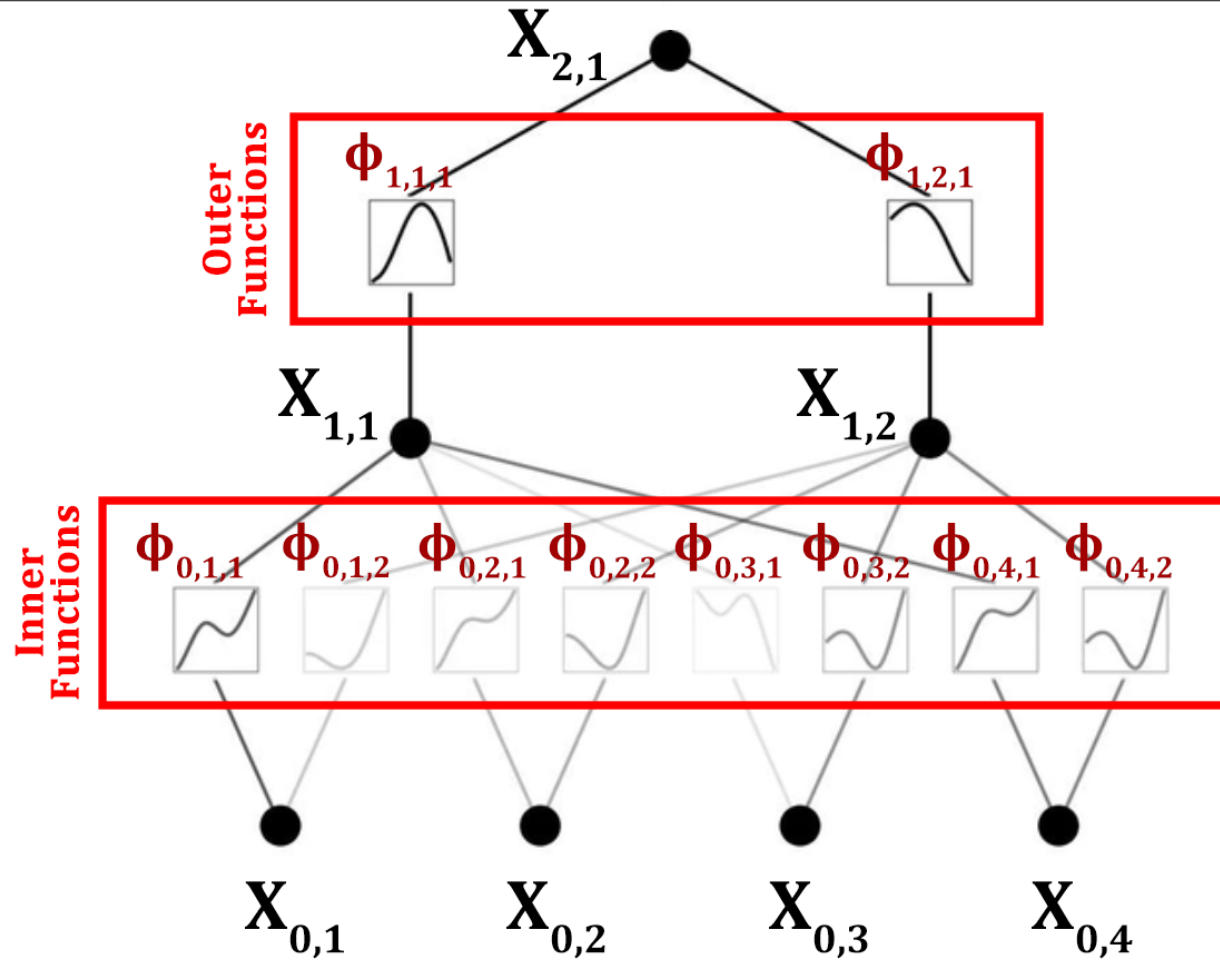
How we intend to proceed



From the 7 papers presented so far, we have drawn the following conclusions:

- ❑ **Missed alarms are weighted more heavily than false alarms**, since the inference is not made too often.
- ❑ **K-Fold, Group K-Fold or even Nested K-Fold Cross Validation can be used instead of the Holdout** method to remove any bias that might arise from the selection of the subset for the test set.
- ❑ Clustering-based EDA can reveal **groups of training samples that are far from the test distribution**.
- ❑ **Non-linear feature combinations can be added** considering that many physical phenomena are described by non-linear polynomials. Then, a *Parallel Coordinates Plot* or a *PCA* can be used to prune less relevant ones.
- ❑ **Environmental features such as np/ng , air density and density altitude can be derived** to normalize the measurements across different operational contexts.
- ❑ The assumption that oat and pa change smoothly over time, together with the consideration that the dataset is recorded sequentially, can be used to **reconstruct the missing temporal information**.
- ❑ A **non-deterministic neural network** should be used to address the request for a PDF output.
- ❑ A **Space-Filling (max-min) strategy** might come in handy **to reduce the size of the training set** in case of a computationally expensive model.
- ❑ Finally, **the principle of Occam's Razor should always be kept in mind**. Almost every paper discussed how overly complex models tended to overfit the training data, with a low test score.

Might KANs help us?



KANs rewrite the original function as a composition of B-form splines.

They proved their superiority over the classic MLPs in several aspects:

- ❑ KANs have learnable activation functions on edges $\Rightarrow$ each "neuronal unit" is by far more expressive than an MLP neuron $\Rightarrow$ a trained network is generally more interpretable.
- ❑ Due to the *locality* property of the splines, KANs can be safely employed in a continual learning scenario, whereas MLP can't.
- ❑ *Symbolic regression* can be easily performed by fitting a function to each spline to reveal potentially useful feature relationships.